

Module 4

Print Settings Fundamentals — Tier 1: Beginner

1. Layer Height: Detail vs. Speed

- 0.28mm — fast, visible layer lines, fine for prototypes/functional parts
- 0.20mm — the standard default, good balance
- 0.12mm — high detail, much slower, best for display models

2. Infill: Strength vs. Weight vs. Material

- 15–20% — standard for most non-structural prints
- 40%+ — for parts that need real mechanical strength
- Infill pattern matters too — grid/honeycomb for general strength, gyroid for balanced strength in all directions

3. Supports: When and How

- Needed for overhangs steeper than roughly 45°
- Two types: normal (touches only the bed) vs. everywhere (touches the model too, more removal work)
- More supports = more material and cleanup — try reorienting the model first to avoid needing them at all

4. Print Speed

Higher speed reduces print time but can hurt quality on complex geometry, stringing, and dimensional accuracy — especially on non-CoreXY printers. Start with your slicer's default profile speed before experimenting.

5. Putting It Together

Goal	Suggested settings
Fast draft/prototype	0.28mm layer, 10-15% infill, high speed

Everyday functional part	0.20mm layer, 20% infill, standard speed
Display/detail piece	0.12mm layer, 15% infill, slower speed

End of Module 4 — Next: Module 5, Troubleshooting Common Failures